

Analysis Report

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GC log file: **gc-2017_08_19-12_53.log.0.current**

Duration: **1 hr 1 min 54 sec**

Long Pause 🔥

Our analysis tells that your application is suffering from long running GC events. **3 GC events took more than 10 seconds.** Long running GCs are unfavourable for application's performance.

Read our recommendations to **[reduce long GC pause \(https://blog.gceasy.io/2016/11/22/reduce-long-gc-pauses/\)](https://blog.gceasy.io/2016/11/22/reduce-long-gc-pauses/)**

⌚ Application waiting for resources ⚠️

It looks like your application is waiting due to lack of compute resources (either CPU or I/O cycles). Serious production applications shouldn't be stranded because of compute resources. In **165 GC event(s)**, 'real' time took more than 'usr' + 'sys' time.

Timestamp	User Time (secs)	Sys Time (secs)	Real Time (secs)
2017-08-19T13:02:03	0.45	0.05	1.48
2017-08-19T13:03:07	0.56	0.05	4.38
2017-08-19T13:05:23	0.46	0.11	1.27
2017-08-19T13:05:29	0.44	0.01	1.93
2017-08-19T13:05:41	0.34	0.02	0.62
2017-08-19T13:05:47	0.29	0.03	0.56
2017-08-19T13:07:33	0.2	0.01	1.04
2017-08-19T13:07:51	0.22	0.01	1.1
2017-08-19T13:08:03	0.35	0.02	1.81
2017-08-19T13:08:09	0.22	0.02	0.59
2017-08-19T13:08:13	0.15	0.01	0.91
2017-08-19T13:08:17	1.1	0.08	4.21
2017-08-19T13:08:33	0.35	0.01	0.64
2017-08-19T13:08:44	0.19	0.01	2.47
2017-08-19T13:09:17	0.27	0.01	0.78
2017-08-19T13:09:18	0.26	0.04	1.63
2017-08-19T13:09:24	0.49	0.02	0.76

2017-08-19T13:09:25	0.55	0.01	1.09
2017-08-19T13:09:29	0.29	0.01	0.62
2017-08-19T13:09:31	0.39	0.06	3.05
2017-08-19T13:09:37	0.24	0.02	0.93
145 more events...			

Read our recommendations to [fix this problem \(https://blog.gceasy.io/2016/12/08/real-time-greater-than-user-and-sys-time/\)](https://blog.gceasy.io/2016/12/08/real-time-greater-than-user-and-sys-time/)

🕒 System Time greater than User Time ⚠️

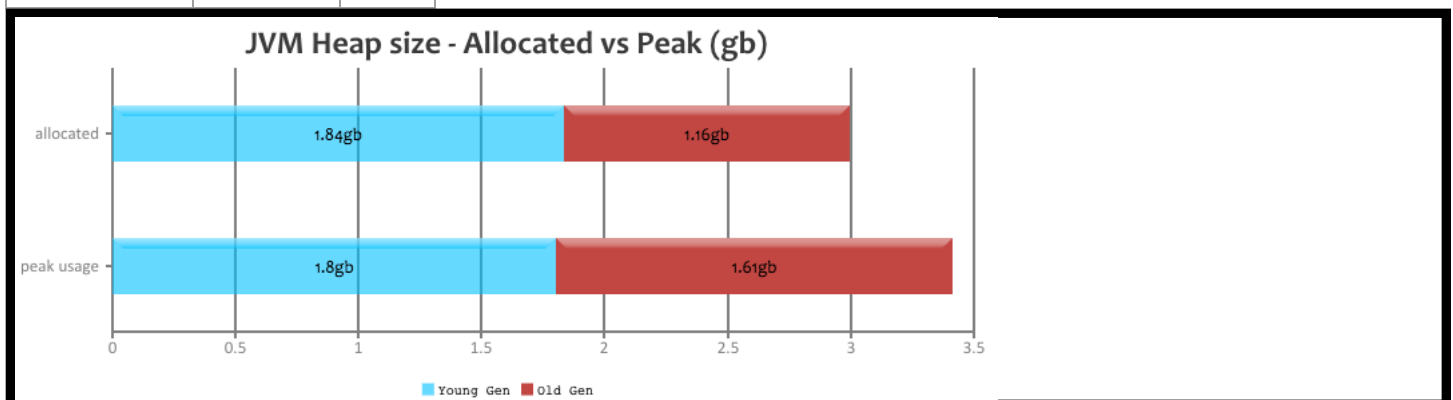
In **5 GC event(s)**, 'sys' time is greater than 'usr' time. It's not a healthy sign.

Timestamp	User Time (secs)	Sys Time (secs)	Real Time (secs)
2017-08-19T12:57:20	0.48	0.86	0.35
2017-08-19T13:00:41	0.29	1.05	0.51
2017-08-19T13:03:04	0.33	0.74	0.87
2017-08-19T13:11:27	0.74	0.95	162.56
2017-08-19T13:37:26	1.21	1.62	90.18

Read our recommendations to [reduce sys time \(https://blog.gceasy.io/2016/12/11/sys-time-greater-than-user-time/\)](https://blog.gceasy.io/2016/12/11/sys-time-greater-than-user-time/)

☰ JVM Heap Size

Generation	Allocated 📄	Peak 📄
Young Generation	1.84 gb	1.8 gb
Old Generation	1.16 gb	1.61 gb
Total	3 gb	2.35 gb



🔍 Key Performance Indicators

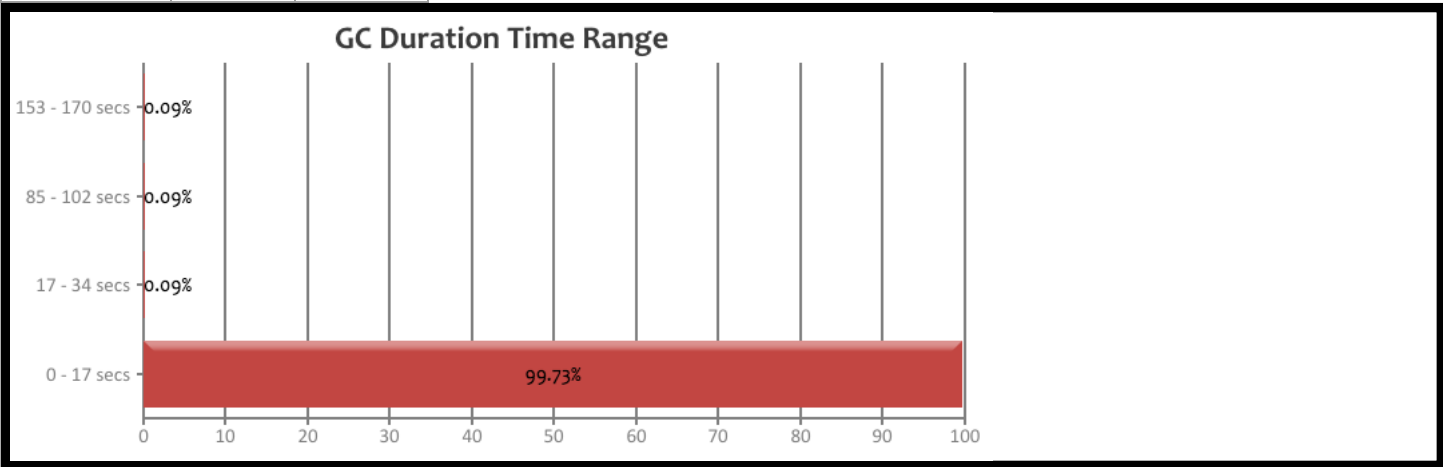
1 Throughput ⓘ : 82.053%

2 Latency:

Avg GC Time ⓘ	586 ms
Max GC Time ⓘ	2 min 42 sec 560 ms

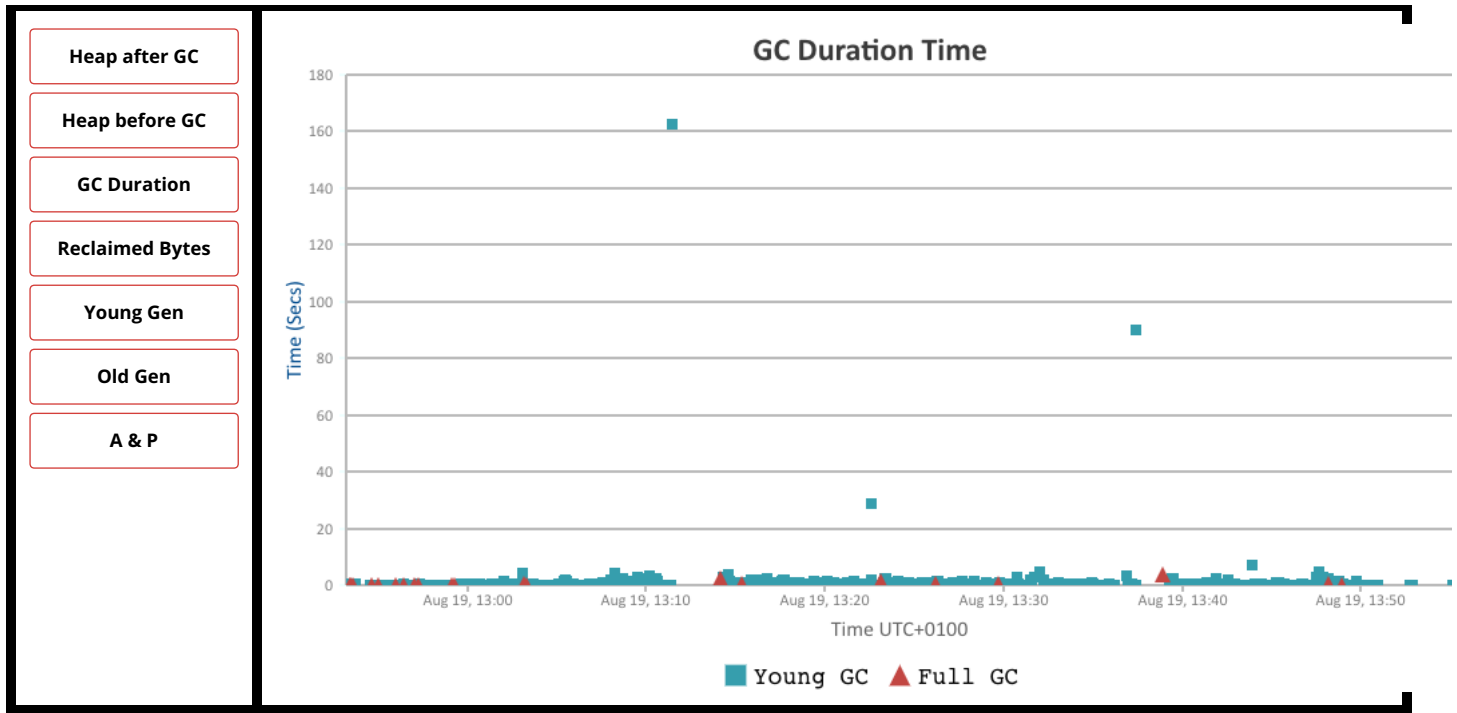
GC Duration Time Range ⓘ:

Duration (secs)	No. of GCs	Percentage
0 - 17	1111	99.731%
17 - 34	1	99.82%
85 - 102	1	99.91%
153 - 170	1	100.0%

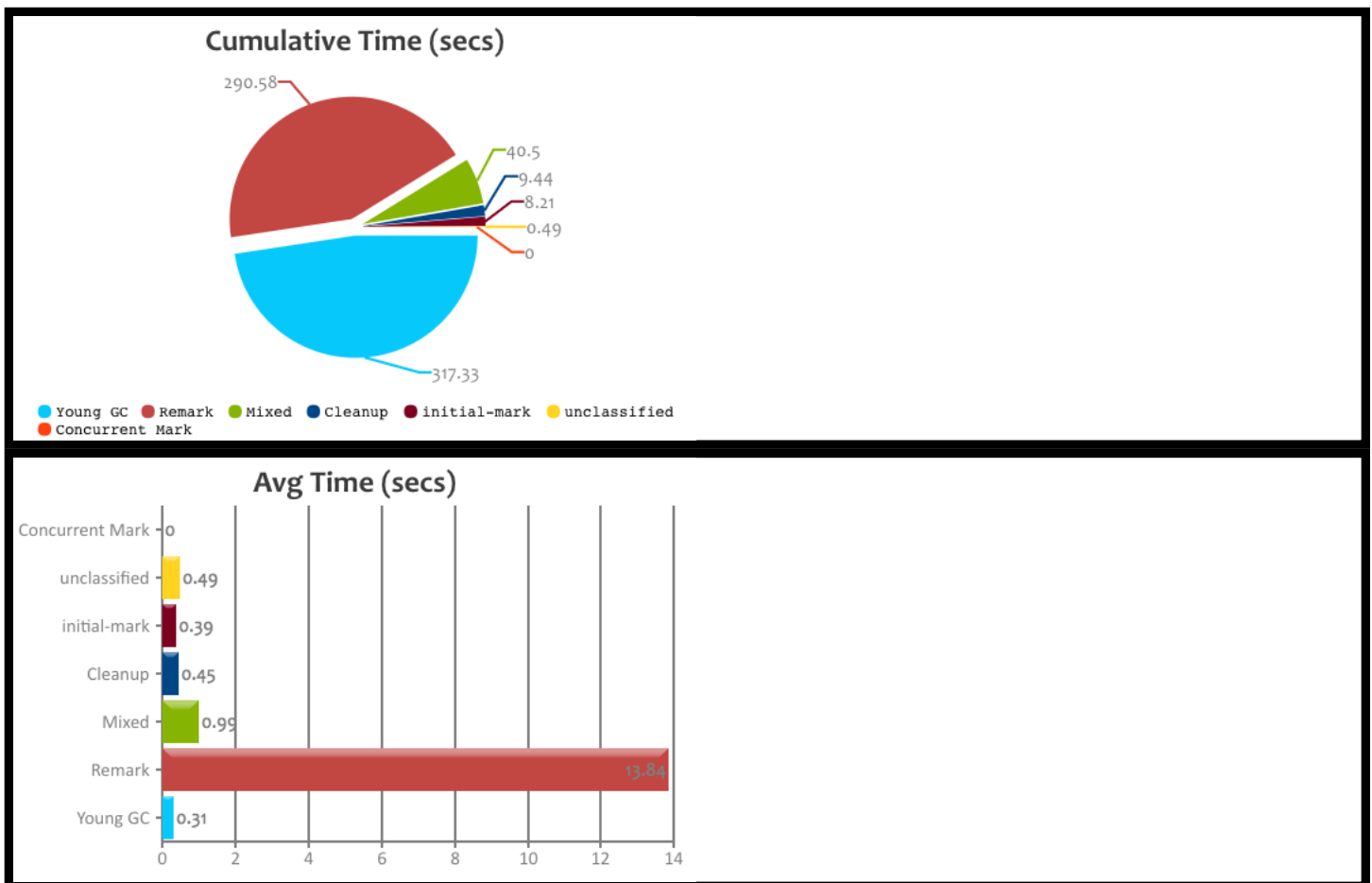


Interactive Graphs

(All graphs are zoomable)



🔍 G1 GC Statistics



	unclassified	Remark	initial-mark	Mixed	Young GC	Concurrent Mark	Cleanup	Total

Count 	1	21	21	41	1012	21	21	1138
Total GC Time 	490 ms	4 min 50 sec 580 ms	8 sec 210 ms	40 sec 500 ms	5 min 17 sec 330 ms	0	9 sec 440 ms	11 min 6 sec 550 ms
Avg GC Time 	490 ms	13 sec 837 ms	391 ms	988 ms	314 ms	0	450 ms	586 ms
Avg Time std dev	0	38 sec 668 ms	519 ms	1 sec 2 ms	552 ms	0	894 ms	5 sec 589 ms
Min/Max Time 	0 / 490 ms	0 / 2 min 42 sec 560 ms	0 / 2 sec 60 ms	0 / 3 sec 990 ms	0 / 7 sec 240 ms	0 / 0	0 / 3 sec 660 ms	0 / 2 min 42 sec 560 ms
Avg Interval Time 	n/a	2 min 46 sec 559 ms	2 min 46 sec 447 ms	1 min 14 sec 18 ms	3 sec 673 ms	2 min 46 sec 559 ms	2 min 46 sec 565 ms	17 sec 681 ms

⚙ Object Stats

(These are good metrics to include in your performance reports)

Total created bytes 	208.02 gb
Total promoted bytes 	4.68 gb
Avg creation rate 	57.35 mb/sec
Avg promotion rate 	1.29 mb/sec

💧 Memory Leak

No major memory leaks.

(**Note:** there are [8 flavours of OutOfMemoryErrors](https://tier1app.files.wordpress.com/2014/12/outofmemoryerror2.pdf) (https://tier1app.files.wordpress.com/2014/12/outofmemoryerror2.pdf). With GC Logs you can diagnose only 5 flavours of them(java heap space, GC overhead limit exceeded, Requested array size exceeds VM limit, Permgen space, Metaspace). So in other words, your application could be still suffering from memory leaks, but need other tools to diagnose them, not just GC Logs.)

⏴ Consecutive Full GC

None.

🕒 Safe Point Duration

(To learn more about SafePoint duration, [click here](https://blog.gceasy.io/2016/12/22/total-time-for-which-application-threads-were-stopped/) (https://blog.gceasy.io/2016/12/22/total-time-for-which-application-threads-were-stopped/))

Not Reported in the log.

❓ GC Causes

(What events caused the GCs, how much time it consumed?)

Cause	Count	Avg Time	Max Time	Total Time	Time %
G1 Evacuation Pause 🕒	1031	316 ms	7 sec 162 ms	5 min 26 sec 116 ms	48.93%
Others	62	n/a	n/a	5 min 23 sec 913 ms	48.6%
GCLocker Initiated GC 🕒	29	319 ms	1 sec 621 ms	9 sec 251 ms	1.39%
G1 Humongous Allocation 🕒	6	746 ms	2 sec 65 ms	4 sec 475 ms	0.67%
Metadata GC Threshold 🕒	9	310 ms	1 sec 87 ms	2 sec 794 ms	0.42%
Total	1137	n/a	n/a	11 min 6 sec 549 ms	100.01%



🔄 Tenuring Summary 🕒

Not reported in the log.

📄 Command Line Flags 🕒

-XX:G1ReservePercent=20 -XX:GCLogFileSize=10485760 -XX:InitialHeapSize=3221225472 -XX:+ManagementServer -XX:MaxGCPauseMillis=200 -XX:MaxHeapSize=3221225472 -XX:NumberOfGCLogFiles=10 -XX:+PrintGC -XX:+PrintGCDateStamps -XX:+PrintGCDetails -XX:+PrintGCTimeStamps -XX:+UseCompressedClassPointers -XX:+UseCompressedOops -XX:+UseG1GC -XX:+UseGCLogFileRotation

🏆 Become a DevOps champion in your organization

(Best practises/tools)

- ✓ Use **fastThread.io** (<https://fastthread.io/>) to analyze thread dumps, core dumps and hs_err_pid files
- ✓ Do proactive Garbage Collection analysis on all your JVMs (not just one or two) using the GC log analysis API 🕒 (<https://blog.gceasy.io/2016/06/18/garbage-collection-log-analysis-api/>)
- ✓ Always rotate the GC log files (<https://blog.gceasy.io/2016/11/15/rotating-gc-log-files/>) to prevent critical garbage collection data loss
- ✓ Use 'Pro' version (<http://gceasy.io/pricing.jsp>) for 10x fast, unlimited, secure usage

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